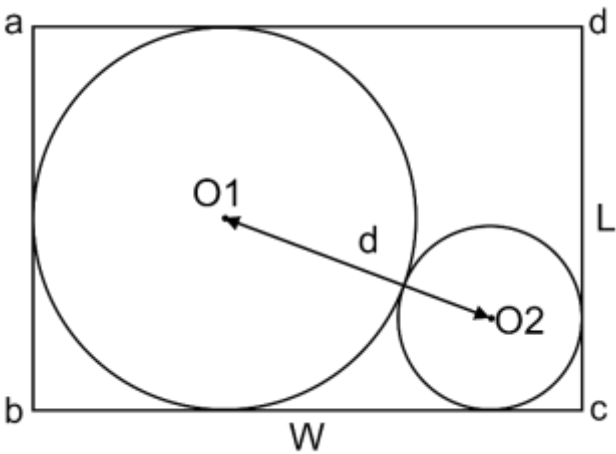


幾何問題

矩形 $abcd$ 裡存在兩個圓 $O1$ 和 $O2$ 且兩圓外切，其中 $O1$ 還與邊長 $\overline{ad}$ 、 $\overline{ab}$ 、 $\overline{bc}$ 相切，而 $O2$ 與邊長 $\overline{dc}$ 、 $\overline{bc}$ 相切，連心線長 $d=\overline{O1O2}$ ，矩形的長和寬分別為 L 和 W ，如下圖所示。試撰寫程式根據 2 個整數：矩形長 L 和連心線長 d ，求該矩形的寬。
「※ $\pi=3.141$ 」



輸入說明：

- 第 1 行輸入欲計算 k 個矩形的寬，其中 $1 \leq k \leq 100$ 。
- 第 2 至第 $1+k$ 行分別輸入 2 個整數，分別為矩形長 L 和連心線 d ，其中 $1 \leq \{L, d\} \leq 1000$ 。

輸出說明：

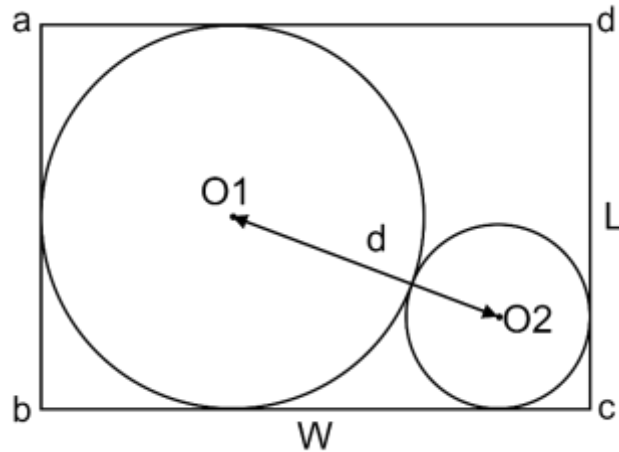
根據每行的矩形長 L 和連心線 d ，計算該矩形的寬 W ，取至四捨五入取至小數點以下第 2 位。

範例說明：

範例 1 輸入	範例 1 輸出
1 6 5	9.90
範例 2 輸入	範例 2 輸出
2 200 170 70 60	337.33 119.16

Geometry Problem

Inside a rectangle $abcd$, there are two externally tangent circles, O_1 and O_2 . Circle O_1 is also tangent to two adjacent sides of the rectangle, and circle O_2 is tangent to the other two adjacent sides. The distance between the centers of the two circles is d . The length and width of the rectangle are L and W , respectively, as shown in the figure below. Write a program to find the width W of the rectangle based on two given integers: the rectangle length L and the distance between the centers d . (Note: $\pi = 3.141$)



Input Description:

The 1st line inputs k , the number of rectangles whose width needs to be calculated, where k is between 1 and 100 inclusive. The 2nd to $(1+k)$ -th lines each input 2 integers, representing the rectangle length L and the center distance d , where both L and d are between 1 and 1000 inclusive.

Output Description:

Based on the rectangle length L and center distance d in each line, calculate the width W of the rectangle, rounded to 2 decimal places.

Example Description:

Example 1

Input

1

6 5

Output

9.90

Example 2

Input

2

200 170

70 60

Output

337.33

119.16